BATTERY INVESTMENT MODEL

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1. Purpose and business question

Naomi has solar panels already which is used to meet electricity power demand first, any remaining demand is purchased from electricity provider, excess solar generated has no financial benefit. A battery could store some of that spare/excess solar power and make it available later, when the house needs electricity. The model estimates how much grid electricity a 12.5 kWh battery could avoid buying, what that could save and whether those savings could justify a $7,000 installation cost.

The analysis uses hourly solar generation and electricity usage from 2020 as a representative year. It models energy flows hour by hour, summarises them by month and projects savings from 2022 to 2041 under two electricity-price inflation scenarios.

- Is the supplied hourly dataset complete and credible enough to model?
- How much electricity would be bought from the grid without a battery?
- How much solar energy is left over after meeting immediate usage?
- How would a 12.5 kWh battery charge and discharge through the year?
- How much grid electricity and money could the battery save?
- Does the investment produce a positive NPV at 6%, and what is its IRR under each price scenario?

The main Business question is will Naomi save enough money by installing a $7,000 battery to justify the investment over its 20 years working life?

2. Data used and initial checks

The source workbook contains a worksheet called “Raw Data”. The analysis uses the timestamp, “Solar electricity generation (kWh)” and “Electricity usage (kWh)” columns. The source data was kept separate from the cleaned working copy so the original observations remain available for audit.

Check	Finding	Action / meaning
Number of hourly records	8,760	Matches 365 days × 24 hours: a full non-leap year.
Timestamp coverage	2020-01-01 00:00 to 2020-12-31 23:00	Covers the full 2020 calendar year.
Missing values in main input columns	None reported initially	No original blanks found in the key inputs before anomaly treatment.
Duplicate rows / timestamps	None reported	No duplicates identified by the checks performed.
Negative solar generation	None reported	No negative solar reading.
Negative electricity usage	5 observations	The five observations were flagged as data-quality anomalies, excluded from the original usage values in the cleaned dataset and replaced using linear interpolation based on neighbouring valid observations.
Extreme electricity usage- outlier	46,000 kWh at 2020-01-12 12:00	Far outside neighbouring hourly readings; likely a recording or unit error, requiring source confirmation. The observation was

		flagged as an extreme outlier, excluded from the original usage values in the cleaned dataset and replaced using linear interpolation based on neighbouring valid readings
Overnight solar reading	13.05 kWh at 2020-04-18 01:00	Unusual for solar generation at 1 a.m.; The reading was flagged for investigation but retained in the primary analysis because its accuracy could not be conclusively determined from the available data. A separate sensitivity analysis was conducted by setting the reading to zero to assess its effect on the model results, the sensitivity analysis showed that the estimated annual battery savings decreased from approximately $595.38 to $594.84, a difference of about $0.54.

Note: The 46,000 kWh value and five negative usage values should be escalated to the data owner or meter provider.

3. Data-quality issues, cleaning and escalation

3.1 Five negative usage values

Timestamp	Original usage (kWh)	Treatment
2020-01-01 17:00	-12.6240	Flagged; set to missing in analysis copy, then interpolated.
2020-02-08 21:00	-2.1330	Flagged; set to missing in analysis copy, then interpolated.
2020-02-08 23:00	-0.2175	Flagged; set to missing in analysis copy, then interpolated.
2020-05-30 17:00	-2.5140	Flagged; set to missing in analysis copy, then interpolated.
2020-09-27 08:00	-1.9770	Flagged; set to missing in analysis copy, then interpolated.

Negative household usage is not consistent with the intended meaning of electricity consumed during an hour. Leaving these values in place could reduce calculated usage and distort grid purchases. The five readings were flagged and excluded from the analysis values.

3.2 Extreme usage of 46,000 kWh

At 12:00 on 12 January 2020, usage was recorded as 46,000 kWh. This is far outside neighbouring readings and would materially distort averages and totals if accepted as a genuine household value. It was flagged as a likely data-entry, unit or meter-export error, set to missing in the working copy, and linearly interpolated using adjacent observations.

Linear interpolation estimates a value between neighbouring observations. It is a temporary modelling treatment, not proof of the true reading. The source value is preserved in the original data and should be replaced if the data owner supplies a verified reading.

3.3 How cleaning was performed

1. Keep the source dataframe unchanged and create a separate working copy for modelling.
2. Preserve the original usage column so the unedited values can be compared with analysis values.
3. Flag usage values below zero and the 46,000 kWh observation as suspect.

4. Replace only those flagged usage values with missing values in the working copy.
5. Use linear interpolation to fill the affected usage observations.
6. Recheck the cleaned usage column for missing and negative values before running the energy model.

After treatment, the cleaned analysis column had zero missing usage values and zero negative usage values. This is the result of cleaning, not evidence that the source file was originally error-free.

3.4 Overnight solar anomaly

Solar generation of 13.05 kWh at 01:00 on 18 April 2020 is unusual because panels normally do not generate meaningful electricity at night. It was not automatically changed in the primary model because the correct value cannot be established from the dataset alone.

Case	Estimated annual savings	Interpretation
Primary model: keep 13.05 kWh reading	$595.38	Main reported result.
Sensitivity: set that solar reading to zero	About $594.84	Savings change by about $0.54, a small effect in this test.

3.5 Escalation log

Issue	Why it matters	Recommended action
46,000 kWh usage on 2020-01-12 at 12:00	Could distort annual usage and savings.	Data owner / meter provider: verify meter record, units and export process.
Five negative usage readings	Could understate usage and grid purchases.	Data owner: confirm whether these reflect resets, export conventions or errors.
13.05 kWh solar at 01:00 on 2020-04-18	Physically unusual for night-time generation.	Solar monitoring provider: check timestamp, timezone and sensor/channel mapping.
Interpolation used for six usage observations	Creates estimated rather than observed values.	Reviewer: approve as temporary treatment and request sensitivity analysis if source verification is unavailable.

Until the source owner responds, results should be described as model estimates based on documented cleaning assumptions, not as verified meter-based savings.

4. Model assumptions

Assumption	How it is used	Potential effect
2020 is representative of future years	2020 hourly usage and solar generation estimate future annual savings.	Weather or household behaviour changes could alter savings.
Battery capacity is 12.5 kWh	Charge is capped at 12.5 and cannot fall below zero.	Usable capacity may be lower in practice.
Battery starts empty	Initial charge at 2020-01-01 00:00 is 0 kWh.	Opening state affects the first hours.
100% charge/discharge efficiency	Every stored kWh is treated as recoverable.	Likely overstates real-world savings.
No power limit, self-discharge, degradation or downtime	Battery is treated as operating consistently.	May overstate the benefit over 20 years.
No export value for unused solar	Surplus energy not used or stored has no financial value.	Consistent with the background; export compensation would change the comparison.

Starting electricity price is $0.17/kWh on 1 Jan 2022	Values avoided grid electricity.	Different tariffs or fixed charges may change savings.
$7,000 cost and 20-year life	Cost is Year 0 outflow; savings projected for 20 years.	No replacement cost or residual value is included.
6% discount rate	Used to calculate NPV.	NPV changes if the required return changes.

5. Methodology and business questions, step by step

Question 1 — Is the data fit for use, and what do hourly patterns show?

I loaded the Excel data into Python and checked its completeness, data types, missing values, duplicate records and summary statistics. The dataset contained 8,760 hourly records covering the whole of 2020, with no missing values, duplicate rows or duplicate timestamps. I then reviewed the minimum and maximum readings and identified five negative electricity usage values and an unusually high reading of 46,000 kWh on 12 January 2020 at 12 p.m. These six usage readings were flagged as anomalies, excluded from the original values in the cleaned dataset and replaced using linear interpolation based on neighboring valid readings. I also identified an unusual solar generation reading of 13.05 kWh at 1 a.m. on 18 April 2020. I retained it in the main analysis and tested its impact separately by setting it to zero; this changed the estimated annual savings by only $0.54, from $595.38 to $594.84. Finally, I plotted the average solar generation and electricity usage for each hour of the day to understand when solar production and electricity demand were highest and identify periods when battery storage could be useful. Overall, the data was suitable for the battery investment analysis after cleaning, although the flagged readings should be verified against the original meter records where possible.

The timestamp was parsed as a date and the hour of day extracted as 0–23. Mean solar generation and mean usage were calculated for each hour across all days in 2020 and plotted together.

Result: solar generation rises through the morning and peaks around 11:00. Usage rises in the evening, particularly around 18:00–20:00. The timing mismatch explains why a battery could help: it can shift some unused daytime solar energy to hours when usage is higher and solar generation is low.

Data issue: the 46,000 kWh usage observation would heavily distort averages if not corrected. The graph should use cleaned data, with the anomaly disclosed.

Question 2 — How much electricity would be bought without a battery?

For each hour, grid purchases are the larger of zero and usage minus solar generation:

$$\text{Grid purchases without battery} = \max(0, \text{usage} - \text{solar generation})$$

When solar meets or exceeds usage, grid purchases for that hour are zero. When usage is higher, the shortfall is bought from the grid.

- Total electricity usage: 18,095.26 kWh consumed during 2020.
- Total solar generation: 9,782.73 kWh generated by the solar system.
- Total grid purchases: 15,639.36 kWh bought from the electricity provider after accounting for solar generation.
- Negative grid purchases: 0, confirming that your formula correctly applies a minimum of zero.

Result: estimated 2020 grid purchases without a battery were 15,639.36 kWh. This result depends on the cleaned usage series.

Question 3 — How much excess solar is available?

$$\text{Excess solar} = \max(0, \text{solar generation} - \text{usage})$$

This is the solar electricity remaining after immediate usage is met. It is the maximum energy available to charge the battery during that hour, before applying the battery capacity limit.

The model uses total solar generation of 9,782.73 kWh and corrected usage of 18,095.26 kWh. Excess solar is not the same as total solar generation; only the amount above immediate usage can charge the battery.

Data issue: the overnight 13.05 kWh solar reading was retained but flagged. The sensitivity run showed a small effect on annual savings.

Question 4 — How was battery charge modelled?

The battery starts at 0 kWh at 00:00 on 1 January 2020. Hour by hour, excess solar charges it up to 12.5 kWh. When usage exceeds solar, stored energy discharges to cover as much of the shortfall as possible. Any remaining shortfall is bought from the grid.

Result and check: battery charge remained between 0.0 and 12.5 kWh. This confirms that the simulation respected the stated minimum and maximum capacity.

Peer-review point: confirm whether reported charge is the end-of-hour level and check the order of operations in each hourly step. The convention must be consistent throughout.

Question 5 — How much grid electricity would be bought with the battery?

When current solar is insufficient, the model uses stored battery energy first. Any remaining shortfall becomes the grid purchase with the battery.

Result: estimated grid purchases with a battery were 12,137.10 kWh in 2020. This is 3,502.26 kWh less than the no-battery case.

Reasonableness check: no hour had higher grid purchases with the battery than without it. That is consistent with the model design.

Question 6 — What are the annual savings at $0.17/kWh?

$$15{,}639.36 - 12{,}137.10 = 3{,}502.26 \text{ kWh saved}$$

$$3{,}502.26 \text{ kWh} \times \$0.17/\text{kWh} \approx \$595.38 \text{ per year}$$

Result: baseline annual saving is approximately $595.38 at the specified 1 January 2022 electricity price, with no discounting. Full precision is retained in calculations and rounded only for display.

Interpretation: this is an estimate under simplified battery assumptions, not a guaranteed bill reduction. Real savings depend on efficiency, tariff structure, household usage and verified meter data.

Question 7 — What does the monthly analysis show?

Hourly values were grouped by calendar month and summed for solar generation, usage, grid purchases without a battery and grid purchases with a battery. The notebook includes a monthly table and charts.

Result: the battery case has lower grid purchases in every month in the model. The benefit varies with solar availability, usage and the amount of excess solar available to charge the battery.

Reasonableness check: monthly totals were reconciled to annual totals and the checks passed. All monthly results inherit the cleaning decisions, including the retained flagged solar reading.

Question 8 — What are the 20-year savings and NPVs?

The projection covers 2022–2041, matching the 20-year battery life. The 2022 saving is the baseline amount at $0.17/kWh. Price inflation begins on 1 January 2023. The $7,000 cost is recorded as a negative Year 0 cash flow, and future savings are discounted at 6%.

Measure	Scenario 1: 4% inflation each year	Scenario 2: inflation rate rises by 0.25 percentage points yearly
Projection period	2022–2041	2022–2041
Total projected savings before discounting	$17,729.27	$21,072.64

Initial investment	$7,000.00	$7,000.00
NPV at 6%	$2,430.74	$3,741.72

Scenario 1 applies 4% inflation each year from 1 January 2023. Scenario 2 applies 4.0% in 2023, 4.25% in 2024, 4.50% in 2025, then increases the annual rate by another 0.25 percentage points each year. The 2022 baseline is not inflated.

Interpretation: both NPVs are positive, so the present value of projected savings exceeds the $7,000 cost under the assumptions. Scenario 2 has the higher NPV because it assumes faster electricity-price inflation.

Review point: the timing matters. Applying inflation to 2022 or starting Scenario 2 at 4.25% in 2023 would overstate savings. The corrected projection uses 2022 as the uninflated baseline.

Question 9 — What are the IRRs?

IRR is the discount rate at which the NPV of the investment cash flows equals zero. The cash-flow series starts with −$7,000 at Year 0, followed by 20 annual savings amounts.

Scenario	IRR	Interpretation
Scenario 1: 4% annual inflation	9.43%	Above the 6% discount rate under the model assumptions.
Scenario 2: inflation rises by 0.25 percentage points yearly	10.72%	Above 6% and higher than Scenario 1 because projected prices rise faster.

Both IRRs exceed 6%, which agrees with the positive NPVs. This is a useful cross-check for these conventional cash flows.

6. Further checks and model controls

Check	Observed result	Why it matters
Rows after cleaning	8,760	No hourly records lost during cleaning.
Missing usage in cleaned analysis column	0	No gaps remained after documented interpolation.
Negative usage in cleaned analysis column	0	Implausible negative readings were removed from analysis values.
Negative grid purchases	0	Grid purchases cannot be negative.
Battery grid purchases greater than no-battery purchases	0 hours	Battery case never increased grid purchases in the hourly comparison.
Battery charge minimum / maximum	0.0 / 12.5 kWh	Battery respects capacity limits.
Monthly-to-annual reconciliation	Passed	Monthly sums agree with annual totals.
Projection length	20 years, 2022–2041	Matches stated battery life.
Year 0 cash flow	−$7,000 in both scenarios	Initial cost included in NPV and IRR.
Solar anomaly sensitivity	$595.38 primary vs about $594.84 with reading set to zero	Small impact in the test performed.

Final conclusion to the business question

Yes, the $7,000 battery investment appears financially worthwhile under the model's assumptions, but it should not be approved for purchase until the key assumptions and real-world costs have been verified.

The analysis estimates that installing a 12.5 kWh battery would allow Naomi to store excess solar energy and use it later, reducing the amount of electricity purchased from the grid.

Key findings

Business measure	Final result
Initial battery investment	$7,000
Estimated annual electricity savings	$595.38
Annual grid electricity avoided	3,502.26 kWh
20-year projected savings — Scenario 1	$17,729.27
20-year projected savings — Scenario 2	$21,072.64
NPV at 6% — Scenario 1	+$2,430.74
NPV at 6% — Scenario 2	+$3,741.72
IRR — Scenario 1	9.43%
IRR — Scenario 2	10.72%

Both scenarios produce a positive Net Present Value (NPV) and both Internal Rates of Return (IRRs) exceed the required 6% discount rate. This means the projected electricity savings are sufficient to recover the initial investment and provide a positive financial return over the battery's assumed 20-year life.

However, these are model-based estimates, not guaranteed returns. The model assumes 100% battery efficiency and excludes battery degradation, maintenance, replacement costs and downtime. These factors could reduce the actual savings. The source data also contains unusual readings that require verification.

Final recommendation

Proceed to a detailed investment review, but do not make the final purchase decision based on the current model alone. Confirm the questionable meter readings, incorporate realistic battery efficiency and operating costs, and test whether the investment remains profitable under less favourable conditions.

The business conclusion is that the battery has potential to deliver a positive financial return, but its actual profitability must be validated before the investment is approved.

7. Limitations and items for approval

- Obtain confirmation of the 46,000 kWh usage observation and five negative readings from the original meter or data owner.
- Confirm the 13.05 kWh solar reading at 01:00 on 18 April 2020, including timezone and sensor/channel mapping.
- Agree whether linear interpolation is acceptable; if not, rerun using an approved alternative and compare results.
- Replace idealised 100% battery efficiency with manufacturer-specified round-trip efficiency and usable capacity.
- Include charge/discharge power limits, degradation, downtime, maintenance and replacement cost where data is available.
- Check whether 2020 solar and consumption patterns are representative of the next 20 years.
- Confirm tariff structure, including fixed charges, export compensation and time-of-use pricing.
- Run sensitivities for discount rate, battery cost, annual energy savings, price inflation and battery life.

8. Recommendation and answer to the business question

Using the cleaned 2020 hourly data and assumptions in this report, the model estimates that a 12.5 kWh battery could reduce grid purchases by 3,502.26 kWh per year and save about $595.38 at $0.17/kWh. Over 20 years, NPV is $2,430.74 under Scenario 1 and $3,741.72 under Scenario 2. IRRs are 9.43% and 10.72%, both above the 6% discount rate.

The figures support taking the investment forward for a more complete review, rather than approving it solely on this model. The conclusion is conditional: source-data anomalies need confirmation, and real-world battery losses, degradation, operating costs and details need to be included before a final purchase decision.

Definitions

Model item	Formula / rule	Meaning
Grid purchases without battery	max(0, usage − solar)	Buy only the shortfall after current solar is used.
Excess solar	max(0, solar − usage)	Solar remaining after current usage is met.
Battery charge	Previous charge adjusted for charge/discharge, bounded from 0 to 12.5 kWh	Stores spare solar and uses stored energy later.
Grid purchases with battery	Remaining shortfall after battery discharge, minimum zero	Grid energy still needed after solar and battery.
Annual energy saved	Grid purchases without battery − grid purchases with battery	Difference in grid electricity purchased.
Annual dollar saving	Energy saved × applicable electricity price	Value of avoided grid electricity.
NPV	−$7,000 + present value of savings discounted at 6%	Whether savings exceed initial cost in today's dollars.
IRR	Rate at which investment cash-flow NPV equals zero	Implied return from projected cash flows.